Politecnico di Milano
School of Industrial and Information Engineering
M.Sc. in High Performance Computing

Final project:
Credit default

Prof. MARAZZINA DANIELE
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Authors:
Jiali Claudio Huang (ID 272686)

1 Introduction

This project addresses the problem of predicting default payments of credit card clients in Taiwan. The goal of this assignment is to build and compare machine learning classifiers to predict whether a client will default on their payment the next month. The methodology explores the implementation of Random Forest and XGBoost classifiers, investigates the effect of feature reduction using Lasso regression, and provides a solution to handle the unbalanced nature of the dataset.

2 Dataset Analysis

The dataset contains 30,000 entries and 25 variables collected from April 2005 to September 2005. The features include demographic factors (such as `SEX, EDUCATION, MARRIAGE, AGE`), historical repayment statuses (`PAY0` to `PAY6`), bill statement amounts (`BILL_AMT1` to `BILL_AMT6`), and previous payment amounts (`PAY_AMT1` to `PAY_AMT6`). During the exploratory data analysis (EDA), it was determined that the dataset contains no missing values. However, the target variable (`default.payment.next.month`) is highly unbalanced.

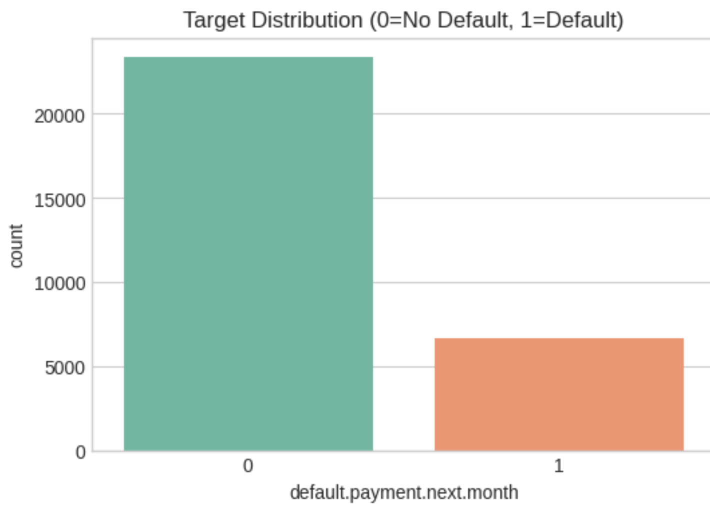

Figure 1: Target distribution

3 Methodology

To classify the default risk, the following machine learning and preprocessing techniques were implemented:

Random Forest: An ensemble classifier. The optimal hyperparameters were determined via GridSearchCV as `max_depth`: 10, `min_samples_leaf`: 4, `min_samples_split`: 3, and `n_estimators`: 300.

XGBoost: A scalable gradient boosting classifier. The best hyperparameters found were `learning_rate`: 0.05, `max_depth`: 3, `n_estimators`: 200, and `subsample`: 0.8.

Lasso Regression (Feature Selection): A linear model with L1 regularization used to reduce the number of features. Lasso successfully reduced the feature space from 23 to 16 key variables, retaining features such as `LIMIT_BAL`, `SEX`, `EDUCATION`, `MARRIAGE`, `AGE`, `PAY_0`, `PAY_2`, `PAY_3`, `PAY_4`, `PAY_6`, `BILL_AMT1`, and several `PAY_AMT` variables.

SMOTE (Synthetic Minority Over-sampling Technique): Proposed as a solution to address the 22.12% class imbalance by generating synthetic examples for the minority default class.

4 Experiment Result

The dataset was split into an 80% training set (24,000 samples) and a 20% test set (6,000 samples) using stratified sampling.

4.1 Baseline Classifiers Comparison

Before feature selection, the models achieved the following baseline performances:

- **Random Forest (Baseline)**: Achieved an AUC of 0.7750 and a default class F1-score of 0.5429.

Table 1: Classification Report: Random Forest (Baseline)

Class/Metric	Precision	Recall	F1-Score	Support
0	0.87	0.85	0.86	4673
1	0.52	0.57	0.54	1327
accuracy			0.79	6000
macro avg	0.70	0.71	0.70	6000
weighted avg	0.80	0.79	0.79	6000

Overall Metrics: AUC = 0.7750, F1 = 0.5429

- **XGBoost (Baseline)**: Achieved an AUC of 0.7788 and a default class F1-score of 0.5352.

Table 2: Classification Report: XGBoost (Baseline)

Class/Metric	Precision	Recall	F1-Score	Support
0	0.88	0.80	0.84	4673
1	0.47	0.62	0.54	1327
accuracy			0.76	6000
macro avg	0.68	0.71	0.69	6000
weighted avg	0.79	0.76	0.77	6000

Overall Metrics: AUC = 0.7788, F1 = 0.5352

While the overall performance of the Random Forest and XGBoost models remains comparable—yielding closely aligned AUC scores of 0.7750 and 0.7788, respectively—XGBoost demonstrates a notable advantage in identifying default instances, improving the minority class recall from 0.57 to 0.62.

4.2 Classifiers with Lasso Feature Selection

After reducing the dataset from 23 to the 16 features selected by Lasso, the models were retrained:

- **Random Forest (Lasso)**: Achieved an AUC of 0.7751 and an F1-score of 0.5388.
- **XGBoost (Lasso)**: Achieved an AUC of 0.7778 and an F1-score of 0.5340. The application of Lasso maintained the predictive power of the models while significantly reducing dimensionality.

Table 3: Classification Report: Random Forest (Lasso)

Class/Metric	Precision	Recall	F1-Score	Support
0	0.87	0.85	0.86	4673
1	0.52	0.56	0.54	1327
accuracy			0.79	6000
macro avg	0.69	0.71	0.70	6000
weighted avg	0.79	0.79	0.79	6000

Overall Metrics: AUC = 0.7751, F1 = 0.5388

Table 4: Classification Report: XGBoost (Lasso)

Class/Metric	Precision	Recall	F1-Score	Support
0	0.88	0.80	0.84	4673
1	0.47	0.61	0.53	1327
accuracy			0.76	6000
macro avg	0.68	0.71	0.69	6000
weighted avg	0.79	0.76	0.77	6000

Overall Metrics: AUC = 0.7778, F1 = 0.5340

By employing Lasso for feature selection, the feature space was successfully reduced from 23 to 16 variables. Although the feature space was reduced, the model's overall performance remained stable. This dimensionality reduction successfully simplifies the model and enhances computational efficiency during training.

4.3 Handling Unbalanced Data and Threshold Optimization

- **XGBoost (Lasso) + SMOTE**: This approach yielded an AUC of 0.7434 and an F1-score of 0.4224. While the overall accuracy decreased to 0.49, the recall for the default class increased significantly to 0.85.

Table 5: Classification Report: XGBoost + SMOTE

Class/Metric	Precision	Recall	F1-Score	Support
0	0.90	0.38	0.54	4673
1	0.28	0.85	0.42	1327
accuracy			0.49	6000
macro avg	0.59	0.62	0.48	6000
weighted avg	0.76	0.49	0.51	6000

Overall Metrics: AUC = 0.7434, F1 = 0.4224

- **Threshold Optimization**: As an alternative optimization, the classification threshold for the XGBoost Lasso model was adjusted from 0.5 to 0.5345, yielding an improved Macro Avg F1-score of 0.7003.

Table 6: Classification Report: XGBoost Lasso + Thresholding

Class/Metric	Precision	Recall	F1-Score	Support
0	0.88	0.84	0.86	4673
1	0.51	0.58	0.54	1327
accuracy			0.78	6000
macro avg	0.69	0.71	0.70	6000
weighted avg	0.79	0.78	0.79	6000

Overall Metrics: AUC = 0.7778, F1 = 0.5426

5 Conclusion

Based on the experiments, the following key insights can be derived:

Table 7: Model Performance Comparison (descending by AUC)

Model	AUC	F1
XGB Baseline	0.778850	0.535165
XGB Lasso	0.777812	0.534031
XGBoost(Lasso) + Thresholding	0.777812	0.542576
RF Lasso	0.775150	0.538823
RF Baseline	0.775043	0.542857
XGBoost(Lasso) + SMOTE	0.743427	0.422414

- **Model Performance**: The XGBoost Baseline model performed the best in terms of overall AUC (0.7788), while the Random Forest Baseline provided the best F1-score (0.5429).
- **Lasso Dimensionality Reduction**: Using Lasso Regression is highly recommended. It reduced the feature set to 16 variables without causing a significant drop in AUC or F1 metrics, leading to a simpler and more robust model.
- **Drivers of Default**: Using SHAP (SHapley Additive exPlanations) values on the XGBoost Lasso model, we can interpret the top drivers of default from the 16 retained features. Historical payment statuses (e.g., PAY_0) and credit limit (LIMIT_BAL) remain the most critical indicators.

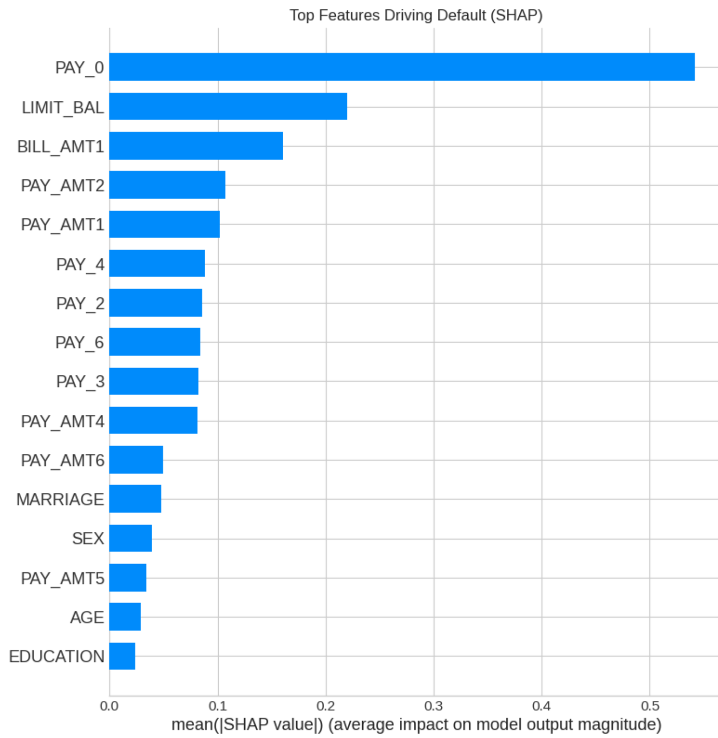

Figure 2: Top features driving default

- **Imbalance Solution**: The dataset is heavily unbalanced. If the primary business goal is risk aversion (identifying as many defaulters as possible), applying SMOTE is the recommended solution as it boosts minority class recall to 85%. For a more balanced performance, implementing Threshold Optimization (threshold = 0.5345) on the XGBoost Lasso model is the optimal strategy.